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30 June 1972
Superseding
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6 September 1967

MILITARY SPECIFICATION

TIMER, PRECISION, BOMBING, 0 - 30 MINUTES

1. SCOPE

1.1 This specification covers the requirements for 28V d-c, integrally lighted, aircraft, precision bombing timers settable from 0 - 30 minutes in 0.1-second increments and having a clock-type display of time-to-go.

1.2 Classification. Timers shall be of the following types, as specified (see 6.2):

Type No.	Lighting	Specification	Voltage
I	White	MIL-L-27160	5V
II	Red	MIL-L-25467	5V
III	White	MIL-L-27160	28V
IV	Red	MIL-L-25467	28V.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416 Plating, Cadmium (Electrodeposited)

Military

MIL-P-116 Preservation, Methods of
MIL-V-173 Varnish, Moisture- and Fungus-Resistant
MIL-E-5400 Electronic Equipment, Aircraft, General Specification for
MIL-L-6085 Lubricating Oil, Instrument, Aircraft, Low Volatility
MIL-S-7742 Screw Threads, Standard Optimum Selected Series: General Specification for

FSC 6645

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MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter; General Specification for
MIL-C-[illegible]	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-701	Preferred and Guidance Lists of Semiconductor Devices
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS3311	Indicator, Elapsed Time, Electrochemical (Mercury Indicating cell)
MS24265	Connectors, Receptacle, Electric-Single Hole Mount, Miniature
MS24266	Connectors, Plug, Electrical - Straight
MS25271	Relay, 10 Amp, 4 PDT, Class B8, Sealed with Solder Hooks
MS33556	Case, Aircraft Instrument, 3-1/4 Inch Size, Standard Dimensions for
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointer, Dial, Standard Design of Aircraft Instrument
MS90386	Indicator, Elapsed Time, Electrochemical (Mercury) 115 Volt AC or 28 Volt DC 50/2400 CPS

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Qualification. The timers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.4).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be ncesssary. Where applicable, varnish in accordance with MIL-V-173 shall be used.

3.3.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to atmospheric and climatic conditions to which the timer may be subjected during storage or normal service.

3.3.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the timer except where magnetic materials are essential.

3.3.4 Nonferrous materials. Nonferrous materials shall be used for all parts of the timer except where ferrous materials are essential.

3.3.5 Toxic and corrosive fumes. Materials used in the construction of the timer shall not liberate gases which combine with the atmosphere to form acid or corrosive alkali, nor liberate toxic fumes when burned.

3.3.6 Protective treatment. When materials are used in the construction of the timer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

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3.4 Design and construction

3.4.1 Design

3.4.1.1 Presentation. The timer shall provide an independent timer signal and shall simultaneously illuminate a TIME OUT light at a preset time-delay interval. The timer setting shall be accomplished by rotating the set knobs on the lower face of the unit. The digital command counters shall indicate in minutes and seconds the value of the input setting. The minute digital counter shall display two figures and the seconds counter shall display three figures of which the third may be set at 0.1 second increments. The counters shall be maintained at the selected positions by a positive detent mechanism. The command numerals shall not count off during timer operation. The time-to-go portion of the presentation shall be a clock-type display with a minute and a second pointer displaying time-to-go (see figure 1).

3.4.1.2 Function. The timer shall be energized by actuating the POWER switch on the face of the timer which will apply a 28V d-c signal to the timer.

3.4.1.3 Sequence. After the desired timing period has been set into the command counters, the START/RESET button shall be depressed once which in turn will automatically set the time-to-go presentation to within 1.0 second of the command counter settings. The time-to-go presentation shall reset to the condition set on the command counters within 3 seconds. When the START/RESET button is depressed a second time, the timing cycle shall immediately begin to function. Upon completion of the timing cycle, an output signal shall be generated and simultaneously the TIME OUT light on the face of the timer shall illuminate. When the START/RESET button is again depressed, the timing sequence shall stop and the timer shall automatically reset to the conditions that are on the command counters. Under no circumstances shall the timer energize the output unless both the POWER and START/RESET switches are actuated and the timer has completed its timing cycle. The START/RESET button shall be white, color No. 37875 of FED-STD-595.

3.4.1.4 Power switch. The POWER switch shall be a three-position rotary type which indicates, in a clockwise direction, OFF, ON, and RELEASE. In the ON position, the timer will be activated and at the end of the timer cycle will actuate the TIME OUT light and provide a signal to an external light. In the RELEASE position, the timer will be activated and at the end of the timing cycle will actuate the TIME OUT light, provide a signal to an external light, and provide an output signal for release to pin 5 of the connector. The release signal shall remain on for a duration of 120 ±10 seconds. The switch shall be maintained at the selected position by a positive detent mechanism. The power switch shall be as shown on figure 1. The switch shall be medium gray, color No. 36231 of FED-STD-595 with a white mark on the face of the switch to indicate its position.

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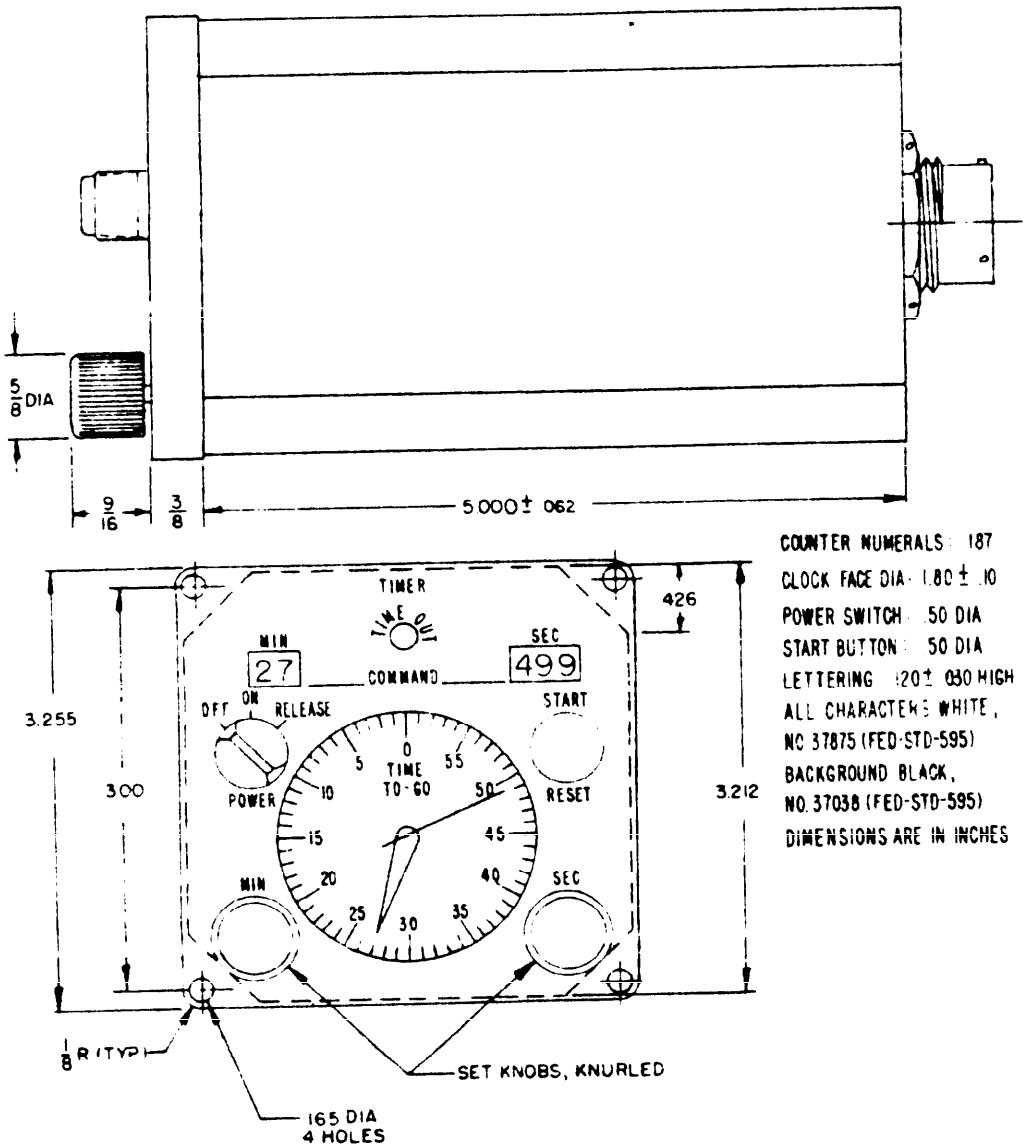


Figure 1. Timer Display

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3.4.1.5 Advisory light. The timer face shall contain one advisory light which shall be a TIME OUT light located at the upper center of the timer face (see figure 1). The light shall have a minimum intensity of 150 ±20 foot-lamberts. The lighting circuit shall be in accordance with figure 2. When the bright-dim control is open, the TIME OUT light shall have an intensity of 1.5 ±0.5 foot-lamberts. The color of the light shall be aviation green in accordance with MIL-C-25050.

3.4.1.5.1 External advisory light connection. The timer shall provide a 28V d-c signal to pin 6 of the connector (see figure 2) each time the timer completes a timing function.

3.4.1.6 Automatic shutoff. The timer mechanism, including both output signals and the TIME OUT light, shall automatically shut off within 120 ±10 seconds after the end of the timing cycle even though the POWER switch remains in the ON or RELEASE position and the 28V d-c power remains in the ON or RELEASE position and the 28V d-c power is continuously applied to the timer. The timer mechanism shall remain off until the START/RESET button is reactivated or the POWER switch is turned off and then on again.

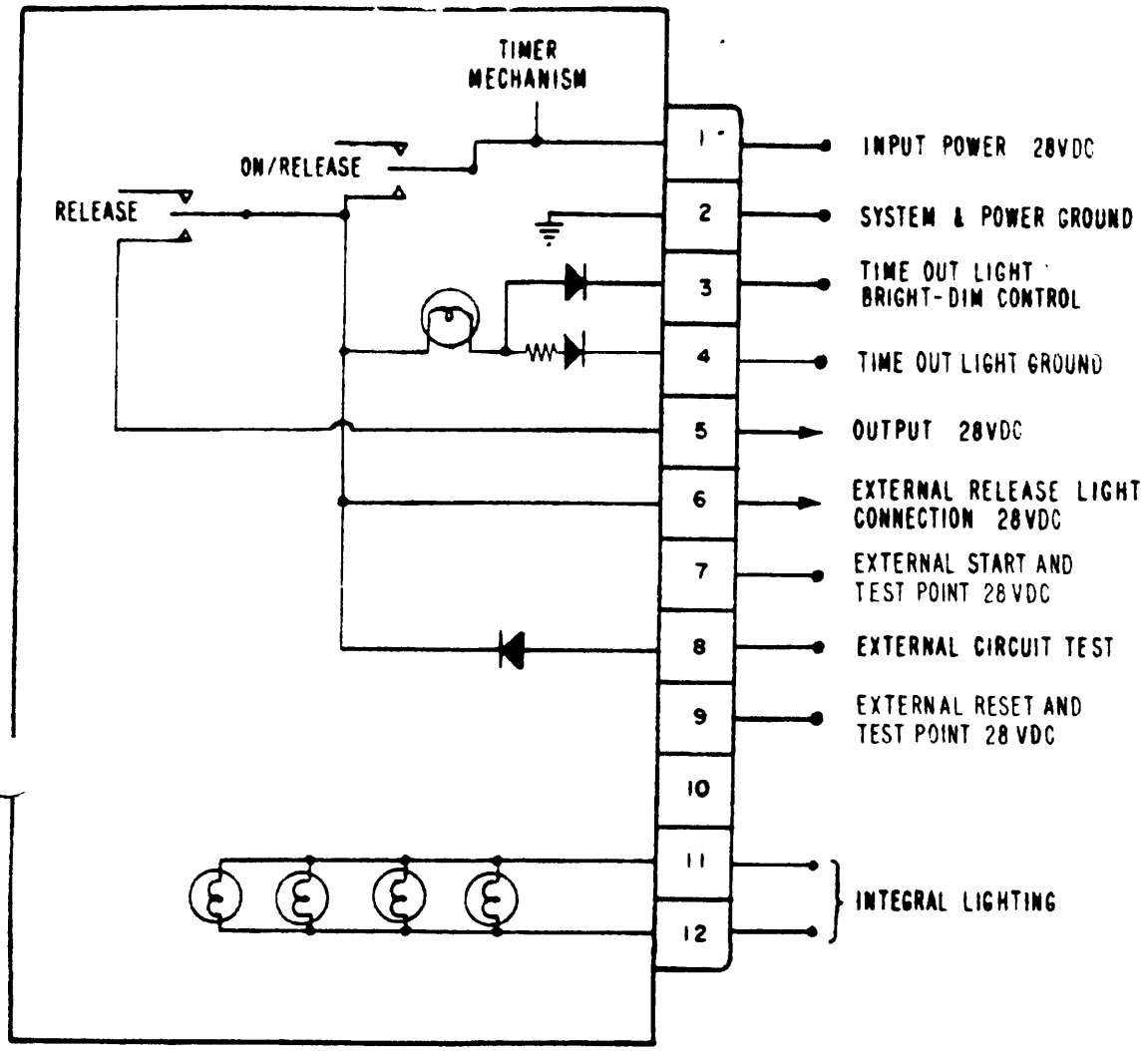
3.4.1.7 Power interruption. When the input power is interrupted for a period up to 0.1 second, the timer shall still maintain the accuracy specified herein.

3.4.1.8 Time setting. The timer shall be provided with two setting knobs as shown on figure 1 for setting the respective minute and second digital command counters. The setting of the minute counters shall be in increments of 1 minute and the second counters shall be in increments of 0.1 second. The torque required to rotate the setting knobs to change tht settings shall be 10 ±3 inch-ounces. A slip clutch shall be incorporated between the setting knobs and the digital counters to prevent overstressing and timer damage. The outer periphery of the setting knobs shall be knurled and the setting knob shafts shall be sealed to prevent moisture from leaking into the timer. The setting knobs shall be medium gray, color No. 36231 of FED-STD-595.

3.4.1.8.1 Knob revolutions. A 2.5 revolution of the setting knob shall be required to set the minutes counter from 0 to 30 minutes and 12 revolutions of the setting knob shall be required to set the seconds counter from 0.0 to 60.0 seconds. Both of the setting knobs shall rotate clockwise for increasing time increments when viewed from the front of the timer.

3.4.1.8.2 Command resetting. The timer shall incorporate a means for re-setting the command counters after the timing function has started which shall not affect the current timing function.

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CONNECTOR

RECEPTACLE: MS24265H12B12P
PLUG: MS24266R12B12S

Figure 2. Circuitry

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3.4.1.9 Timing range. The total range of the timer shall be from 0 to 30 minutes for the minutes counter and from 0.0 to 59.9 seconds for the seconds counter. Thus, the timing limit will be 30 minutes 59.9 seconds.

3.4.1.9.1 Accuracy. After a 20-second warmup period, the timer shall maintain an accuracy of at least ±0.10 percent of the command counter setting with a lower limit (below 100 seconds) accuracy of at least 0.1 second throughout the performance envelope specified herein. The timer scale error shall be within the tolerances specified in table I. These tolerances shall be applicable to all timer settings of 0.2 second and above.

TABLE I. Scale Error

MINUTES	SECONDS	SCALE ERROR TOLERANCES ±SECONDS
* 5	0	0.300
*10	0	0.600
20	0	1.200
25	0	1.500
*30	0	1.800
*17	0	1.020
* 7	0	0.420
0	* 5	0.100
0	*10	0.100
0	20	0.100
0	*30	0.100
0	40	0.100
0	*50	0.100
0	59.9	0.100
* 2	* 2	0.122
5	5	0.305
10	10	0.610
*15	*15	0.915
20	20	1.220
*25	*25	1.525

*Abbreviated test schedule

3.4.1.10 Calibration adjustment. The timer may incorporate a calibration adjustment on the rear of the case. Calibration shall be maintained within the accuracy tolerance specified herein for a period of 2,500 hours of operation.

3.4.1.11 External circuitry. The timer shall be designed to operate with external electrical circuitry as shown on figure 2.

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3.4.1.12 Operating voltage. Except for the lighting circuit, the timer shall be designed to operate from aircraft electrical power systems having characteristics in accordance with MIL-STD-704. The accuracy specified herein shall apply for all power input voltages from 25.0 to 29.0V dc.

3.4.1.12.1 Current ratings. The current ratings of the various timer components shall be as follows:

a. Timer mechanism - not to exceed 0.17 amp

b. Clutch - not to exceed 0.17 amp for each operation

c. Switches and relays - shall be capable of handling at least 1.0 amp inductive load at 28V dc

d. When operating on rated power, the timer shall not draw more than 600 milliamps with the bulbs unlit.

Note: In a solid-state design, the total of a and b shall not exceed 0.34 amp.

3.4.1.12.2 Output voltage tolerance. The output signal voltage from pins of the timer connector shall be within 1 percent of their input voltages under loads up to 1.0 amp.

3.4.1.13 Lighting. Integral lighting shall be in accordance with MIL-L-25467 or MIL-L-27160, as applicable (see 1.2).

3.4.1.13.1 Prime standard. The prime standard lighting reference instrument shall be prepared as follows: A minimum of 20 areas, covering the entire face of the timer, shall be measured for brightness, color, and stray light as specified in the applicable lighting specification. The reference prime standard shall be approved by the procuring activity.

3.4.1.14 Reliability. The timer shall have a minimum acceptable mean-time-between-failures (MTBF) of 1,000 hours demonstrated to a 90 percent confidence level and a discrimination ratio of 1.5.

3.4.1.14.1 Longevity. The timer shall have an operating life span (equipment longevity) of not less than 2,500 hours before wearout failures occur or the equipment consistently fails to meet the specified MTBF index. Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the procuring activity. Life requirements of lamps shall be in accordance with applicable specifications approved by the procuring activity.

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3.4.1.15 Dial. The timer dial shall conform to figure 1. The numerals and letters shall conform to MS33558.

3.4.1.15.1 Dial to cover glass distance. The distance between the digital counters and the inside of the cover glass shall not exceed 0.100 inch.

3.4.1.15.2 Readability. With the timer placed at eye level or lower, and when viewed from any direction, the counter numerals and the dial shall be clearly readable within a 45° angle from a line perpendicular to the timer face. No light leakage shall be observed within this cone.

3.4.1.16 Pointers. The pointers shall conform to figure 1 and MS33585.

3.4.1.17 Case. The case shall conform to MS33556 and figure 1 and shall be so sealed as to prevent moisture and dust from entering the case. Sealing shall be such that it will not be dependent upon materials that might be affected by the action of any atmospheric conditions to which the case may be subjected. The bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in flat black, color No. 37038 of FED-STD-595. The remainder of the case may be finished in black and shall be adequately protected to satisfactorily withstand the environmental conditions specified herein. The front edge of the case shall not extend beyond the outer surface of the cover glass by more than 1/32 inch. The case shall be so designed that the internal mechanism may be removed from the case and replaced without the use of special tools.

3.4.1.18 Elapsed time indicator. Each timer shall include one electrochemical elapsed time indicator conforming to MS3311-1 or MS90386-4. The indicator shall be electrically connected to the timer circuitry, shall be energized when the timer is operating, and shall be so located that it will be easily readable when the timer cover is removed.

3.4.2 Construction. The timer shall be so constructed that adjustments and repairs can be easily made by personnel of overhaul bases without the use of special tools.

3.4.2.1 Transistors and diodes. Transistors and diodes used in the construction of the timer shall be of the silicon type designed for continuous operation in a 100°C ambient temperature and conforming to MIL-STD-701. Circuit design shall be such that no electronic component will be operated outside its rated characteristics with power input as specified herein. No electronic device shall break down when excited by transients as specified in MIL-STD-704.

3.4.2.2 Electronic components. Electronic components and the application thereof shall conform to MIL-E-5400. Nonstandard parts shall be approved by the procuring activity prior to fabrication of production items.

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3.5 Performance. The timer shall operate satisfactorily and shall meet the requirements specified herein when subjected to the following conditions:

a. Temperatures - Temperatures ranging from -54° to +71°C

b. Vibration - Vibration at frequencies between 5 and 500 Hertz and at applied double amplitudes ranging up to 0.4 inch maximum

c. Humidity - Relative humidity up to 100 percent

d. Fungus - Exposure to fungus for 28 days as specified in section 4

e. Sand and dust - Exposure to sand and dust as encountered in desert areas

f. Salt fog - Exposure to salt fog for 48 hours

g. Altitudes - Altitudes ranging from sea level to 72,000 feet.

3.5.1 Dielectric strength. The timer shall withstand a potential of 500V rms commercial frequency for a period of 1 minute applied between the connector pins and the case and between any connector pins that are isolated, circuitry wise, from any other pins of the connector.

3.5.2 Electromagnetic interference. The timer shall meet the electromagnetic interference requirements of MIL-STD-461 for class A1 equipment.

3.5.3 Operating position. The timer shall be capable of operating in any position.

3.5.4 Explosive decompression. The timer shall meet the performance requirements specified herein immediately after the unit has been subjected to condition I of an explosive decompression test in accordance with 4.6.23. It shall not deform nor disintegrate in a manner that could cause injury to a crew member or otherwise jeopardize aircraft safety when subjected to condition II of 4.6.23.

3.5.5 Magnetic effect. The timer, operating or nonoperating, shall not create a mangetic field with such intensity as to deflect a short bar magnetic compass more than 2°.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

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3.7 Cover glass. The cover glass over the timer shall be clear and free from flaws that would interfere with normal reading of the timer. The cover glass shall be secured in place and shall be leakproof, and shall not be fluorescent when exposed to ultraviolet light or rays in the 363 millimicron region. All reflecting glass surfaces shall be provided with a reflection-reducing coating which shall conform to MIL-C-14806 and which shall meet the environmental conditions specified herein.

3.8 Connector. The connector shall be as shown on figure 2.

3.9 Finishes and protective coatings

3.9.1 Exterior. Unless otherwise specified, the exterior of the timer shall be finished in durable dull black, color No. 37038 of FED-STD-595. When aluminum is used, it may be finished with a black anodized coating.

3.9.2 Plating. Plating shall not be required on normally lubricated parts, such as gears, bearings, and shafts (fabricated from brass), bronze, or corrosive-resistant steel.

3.9.3 Cadmium-plated parts. Cadmium-plated parts shall be in accordance with QQ-P-416, and of a type and class necessary for the protection required.

3.9.4 Aluminum alloy parts. Aluminum alloy parts shall be covered with an anodic film in accordance with MIL-A-8625, except that small holes, straight threads, and case inserts need not be anodized.

3.10 Weight. The weight of the timer shall not exceed 3.125 pounds.

3.11 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the timer, they shall be held or limited accordingly.

3.12 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742 or MIL-S-8879.

3.13 Lubrication. Lubrication used in the timer shall conform to MIL-L-6085.

3.14 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.14.1 Schematic. An electrical schematic showing the function of each connector pin shall be permanently attached to the body of the timer. An approved metal decalcomania may be used provided it can not be peeled back with the fingernail.

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3.14.2 Nameplate. A nameplate shall be permanently attached to the body of the timer. Where applicable, the information shall conform to MIL-STD-130. An approved metal decalcomania may be used provided it can not be peeled back with the fingernail. This nameplate may be combined with the electrical schematic specified in 3.14.1.

3.15 Workmanship. The timer, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges, accuracy of dimensions, radii of fillets, and marking of parts and assemblies, thoroughness of soldering, welding, brazing, painting, wiring, and riveting, alignment of parts, and tightness of assembly screws and bolts.

3.15.1 Gears. Gear assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Where required for accurate adjustments, gear assemblies shall be free from backlash.

3.15.2 Cleaning. After final assembly, the timer shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, resin flash that may crumble, and other foreign materials shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of timers shall be classified as follows:

a. Qualification tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25°C). When tests are made with

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atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Attitude. Unless otherwise specified, the timer shall be tested in such a position that the lighting plate in the face of the timer will be vertical.

4.3.3 Test voltage. Unless otherwise specified, the timer shall be tested with a 28V d-c power supply, except for the lighting circuit which shall be tested with a nominal 5V supply.

4.3.4 Vibration. Unless otherwise specified, the timer shall not be subjected to any form of vibration or tapping during the performance of the tests specified herein.

4.3.5 Test sequence. All tests shall be run in the order specified herein.

4.4 Qualification testing

4.4.1 Test samples. The test samples shall consist of at least 10 timers representative of the production equipment. Four of the samples shall be subjected to all the tests except reliability and longevity and the other six samples shall be subjected to the reliability and longevity tests only. The samples shall be identified with the manufacturer's part number and such other information as required by the letter of authorization. The contractor shall submit to the qualifying activity all samples upon which qualification tests have been performed.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.2.1).

4.4.3 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests. Each timer shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Operational checks

c. Visual lighting check

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d. Setting knob torque

e. Scale error

f. Voltage drop.

4.5.2 Sampling tests

4.5.2.1 Sampling plan A. Two timers shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as described under 4.6:

a. Individual tests

b. Voltage variation

c. High temperature

d. Low temperature

e. Vibration error

f. Integral lighting.

4.5.2.2 Rejection and retest. When one timer selected from a production run fails to meet the specification, no timers still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.2.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Examination of product. The timer shall be inspected to determine compliance with the requirements specified herein with respect to materials, marking, workmanship, and general conformance to figures 1 and 2.

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4.6.2 Operational checks

4.6.2.1 Mechanical operation. The timer shall be checked for mechanical operation by manipulating the setting knobs sufficiently to operate the timer readout from the lower to the upper stop. Travel below or above the specified stop location shall not be sufficient to allow the timer readout to detent on any timer setting below or above the specified limits (0 and 30 for the minutes portion of the timer and 0.0 and 59.9 for the seconds portion of the timer). Detent action shall be observed during this manipulation to insure that the digits of the timer stop in satisfactory readable alignment at each detent. The setting knobs shall be driven against the stops for three full turns of the knob at both ends of the scale to determine that the slip clutch operates properly. The time-to-go portion of the timer shall be checked to see if the START/RESET button operates properly and to determine that the clock pointers align properly and display the time selected on the command counters within 1.0 second.

4.6.2.2 Electrical operation. The timer shall be connected to a 28V d-c power supply making sure that the POWER switch is in the OFF position initially. The initial timing check shall be made by setting in the maximum readout on both command counters 30 minutes 59.9 seconds. The timer warmup shall also be checked on the first run as follows: The POWER switch shall be turned to RELEASE and the START/RESET button shall be depressed once whereas the time-to-go presentation shall align itself to the command counter setting. Exactly 20 seconds after the POWER switch is initiated, the START/RESET button shall be depressed again to start the timer. During timer operation, the time-to-go presentation shall count down and display the time-to-go within an accuracy of 1.0 second. The time which elapses between the initiation of the START switch and the initiation of the output signal and illumination of the TIME OUT light shall be measured and recorded. A similar run shall be made using a command reading of 15 minutes 30 seconds and similarly a third run shall be made with a command setting of 0 minutes 10.0 seconds. At no time shall the output signal and light timing differ from the command settings by more than the prescribed accuracy (see 3.4.1.9.1). The above timing procedure for the three sets of readings shall be repeated except that the input power signal shall first be interrupted for 100 milliseconds midway through the timing cycle. At the end of this test, the timer shall be checked for compliance with the automatic shutoff requirements.

4.6.2.3 Setting knob operation. A 0-minute 0.0-second time reading shall be set on the timer counters. The setting knob shall then be rotated so that both counters are at full scale, 30 minutes 59.9 seconds. The action of the setting knobs shall be observed for compliance with the following:

a. Rotation of the setting knobs in a clockwise direction when viewed from the front shall produce an increase in the timing interval on the digital counters

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b. Increasing or decreasing of the counter setting by 0.1 second for each detent for the seconds counter and 1 minute for each detent for the minutes counter

c. 2.5 turns to move the minute counter from 0 to 30 minutes and 12 turns to move the second counter from 0.0 to 60.0 seconds

d. Slip clutch action between setting knobs and the counter mechanisms when attempting to set beyond the upper and lower timer limits.

4.6.2.4 Time-to-go display operation. The time-to-go display portion shall be operated and observed for compliance with the following:

a. The START/RESET button shall be depressed 10 times to determine that the first, third, fifth, et cetera depressions reset the time-to-go display to the reading on the command counters. Different command settings shall be used for each reset action

b. In the 10 depressions of the START/RESET button (see subparagraph a), the alternate depressions (2, 4, 6, et cetera) shall start the timer mechanism

c. In the reset actions (see subparagraph a), time-to-go display shall reset to within 1.0 second of the command setting.

4.6.3 Visual lighting check. The lighting circuit of the timer shall be connected to a power source of rated lighting-system voltage and compared visually with a lighting standard that meets the lighting requirements specified herein. There shall be no noticeable lighting difference between the two timers.

4.6.4 Setting knob torque. The amount of torque required to turn the setting knobs shall be measured and shall be 10 ±3 inch-ounces.

4.6.5 Scale error. The timer shall be connected to the test equipment with a load applied to the output pins as specified. The timing intervals shall then be set into the timer for each set of test points listed in table I; the time-out signals shall be within the tolerance specified therein. The time which elapses between the closing of the START switch and the initiation of outputs shall be measured with the equipment having an accuracy of at least one order of magnitude greater than the accuracy specified in table I.

4.6.6 Voltage drop. Separate 1.0-amp loads shall be connected across pins 2 and 5 and across pins 2 and 6. The timer shall then be caused to produce an output signal. The voltage drop across pins 2 and 5 and across pins 2 and 6 shall be measured and shall be within 1 percent of the input voltage across pins 1 and 2.

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4.6.7 Voltage variation. The timer shall be connected to a nominal 28V d-c power source and subjected to the test specified in 4.6.2.2, first with 25.0V d-c input to the timer and then with a 29.0V input. The timing accuracies specified therein shall apply.

4.6.8 High temperature. The timer shall be subjected to a high temperature test in accordance with method 501 of MIL-STD-810. Near the end of the test, the timer shall be subjected to 3 cycles of operation at both midscale and full-scale settings. The tolerances shall not exceed those specified in 3.4.1.9.1. Upon completion of this test, the timer shall meet the test specified in 4.6.5.

4.6.9 Low temperature. The timer shall be subjected to a low temperature test in accordance with method 502 of MIL-STD-810. The test shall consist of 3 cycles of operation at both midscale and full-scale timer settings at the -54°C temperature. Tolerances shall not exceed those specified in 3.4.1.9.1. Upon completion of this test, the timer shall meet the test specified in 4.6.5.

4.6.10 Vibration error. The timer shall be subjected to a vibration error test in accordance with MIL-STD-810, method 514, category (b), curve C as specified in the vibration test selection chart. This test shall be conducted while the timer is being subjected to 3 cycles of operation at midscale and full-scale settings. The tolerances shall not exceed those specified in 3.4.1.9.1. Upon completion of this test, the timer shall meet the test specified in 4.6.5.

4.6.11 Vibration failure. The timer shall be subjected to a vibration failure test in accordance with MIL-STD-810, method 514, category (b), curve C as specified in the vibration test selection chart. At the conclusion of this test, the timer shall be subjected to 3 cycles of operation at midscale and full-scale settings. The tolerances shall not exceed those specified in 3.4.1.9.1. Upon completion of this test, the timer shall meet the test specified in 4.6.5.

4.6.12 Integral lighting. Lighting tests shall be conducted on all qualification test units in accordance with MIL-L-25467 for red lighting and MIL-L-27160 for white lighting, as applicable. Lighting acceptability of production timers shall be based on visual comparison with the prime standard.

4.6.12.1 Visual comparison. Production timers shall be checked by comparison with the prime standard defined in 3.4.1.13.1. The prime standard and the instrument under test shall be placed side by side at eye level. Both units shall be connected to a common variable source (0-5V). The brightness of the two units shall be compared with 2.85V applied to the standard and 2.85V applied to the unit under test. If the brightness of the two units appears the same visually, the unit under test shall be considered acceptable. If any area of the instrument under test appears visually to differ from the same area of the

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prime standard, it shall then be subjected to all display tests specified in MIL-L-25467 or MIL-L-27160, as applicable, and sufficient data shall be taken to determine compliance or noncompliance with the display requirements therein.

4.6.12.2 Visual color comparison. A voltage of 4.5V shall be applied to the prime standard and timer being tested. The two timers shall be compared and shall visually appear identical in color. If the color of the two units appears the same visually, the unit under test shall be considered acceptable. If any area of the unit under test appears visually to differ from the same area of the prime standard, the unit shall be subjected to the color tests specified in MIL-L-25467 or MIL-L-27160, as applicable, and sufficient data taken to determine compliance or noncompliance with the color requirements therein.

4.6.12.3 Advisory light. The TIME OUT light shall be activated and its color and intensity measured and recorded. The TIME OUT light intensity shall be as specified in 3.4.1.5 and shall illuminate the precise moment the timing cycle runs out.

4.6.13 Magnetic effect. The timer, not operating, shall be rotated in a vertical plane about a short-bar magnetic compass with the nearest part of the timer 5-1/2 inches from the center of the compass. The vertical plane shall be oriented east-west through the center of the compass and the face of the timer shall remain in this plane. Starting directly under the compass, the timer shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the timer shall be rotated on its own horizontal axis until it is in the normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 2°. This test shall be repeated with the timer operating at rated voltage.

4.6.14 Electromagnetic interference. The timer shall meet the electromagnetic interference requirements of MIL-STD-462 for class A1 equipment.

4.6.15 Power interruption. The timer shall be subjected to a scale error test as specified in 4.6.5 at room temperature. At each test station, the input power signal shall be momentarily interrupted midway through the timing cycle. The signal interrutpion time shall be varied between test points and shall include, but not exceed, 100 milliseconds. The tolerances listed in table I shall apply for all test stations.

4.6.16 Reflectance. The reflectance of all reflecting glass surfaces shall be measured and the tolerances shall not exceed those specified in MIL-C-14806.

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4.6.17 Dielectric strength. The timer shall be placed in a chamber having a pressure of 1.67 inches Hg absolute (65,000 feet) and a potential of 500V commercial frequency applied for a period of 1 minute across the connector pins or pins and ground. There shall be no evidence of breakdown. The case grounds may be removed for this test.

4.6.18 Humidity. The timer shall be subjected to a humidity test in accordance with MIL-STD-810, method 507, procedure I. After completion of this test, the timer shall be subjected to and shall meet the tests specified in 4.6.2.1, 4.6.2.2, 4.6.5, and 4.6.6 except the abbreviated test schedule shall apply.

4.6.19 Dust. The timer shall be subjected to a dust test in accordance with MIL-STD-810, method 510, procedure I. The timer shall then meet the tests specified in 4.6.2.1, 4.6.2.2, 4.6.5, and 4.6.6, except the abbreviated test schedule shall apply.

4.6.20 Altitude. The timer shall be placed in a chamber at an absolute pressure of 1.15 inches Hg (72,000 feet) and tested as specified in MIL-STD-810, method 500, procedure II. While at the specified pressure, the timer shall meet the test specified in 4.6.2.2. The rate of pressure change shall be 500 ±50 feet per minute.

4.6.21 Fungus. The timer shall be subjected to a fungus test in accordance with MIL-STD-810, method 508, procedure I. At the conclusion of this test, the timer shall be removed from the chamber and subjected to the test specified in 4.6.2.2.

4.6.22 Salt fog. The timer shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509, procedure I. At the completion of this test, the timer shall be removed from the chamber and subjected to the tests specified in 4.6.2.2 and 4.6.5, except the abbreviated test schedule shall apply.

4.6.23 Explosive decompression. The timer shall be subjected to an explosive decompression test in which the surrounding air pressure varies with time in accordance with conditions I and II of figure 3. The requirements specified in 3.5.4 shall be met. The timer shall then meet the tests specified in 4.6.2.1, 4.6.2.2, 4.6.5, and 4.6.6, except that the abbreviated test schedule shall apply.

4.6.24 Setting knob and counter cycling. The minute set knob shall be continuously cycled from 2 to 28 minutes ±1 minute and return to 2 minutes, and the seconds set knob shall be continuously cycled from 01.0 second to 59.0 seconds ±0.5 second and return to 01.0 second. This excursion shall be considered one cycle of operation. The total cycling period shall be a minimum of 2,500 cycles for each function and shall be conducted at a rate of 2 ±0.5 cpm. The last

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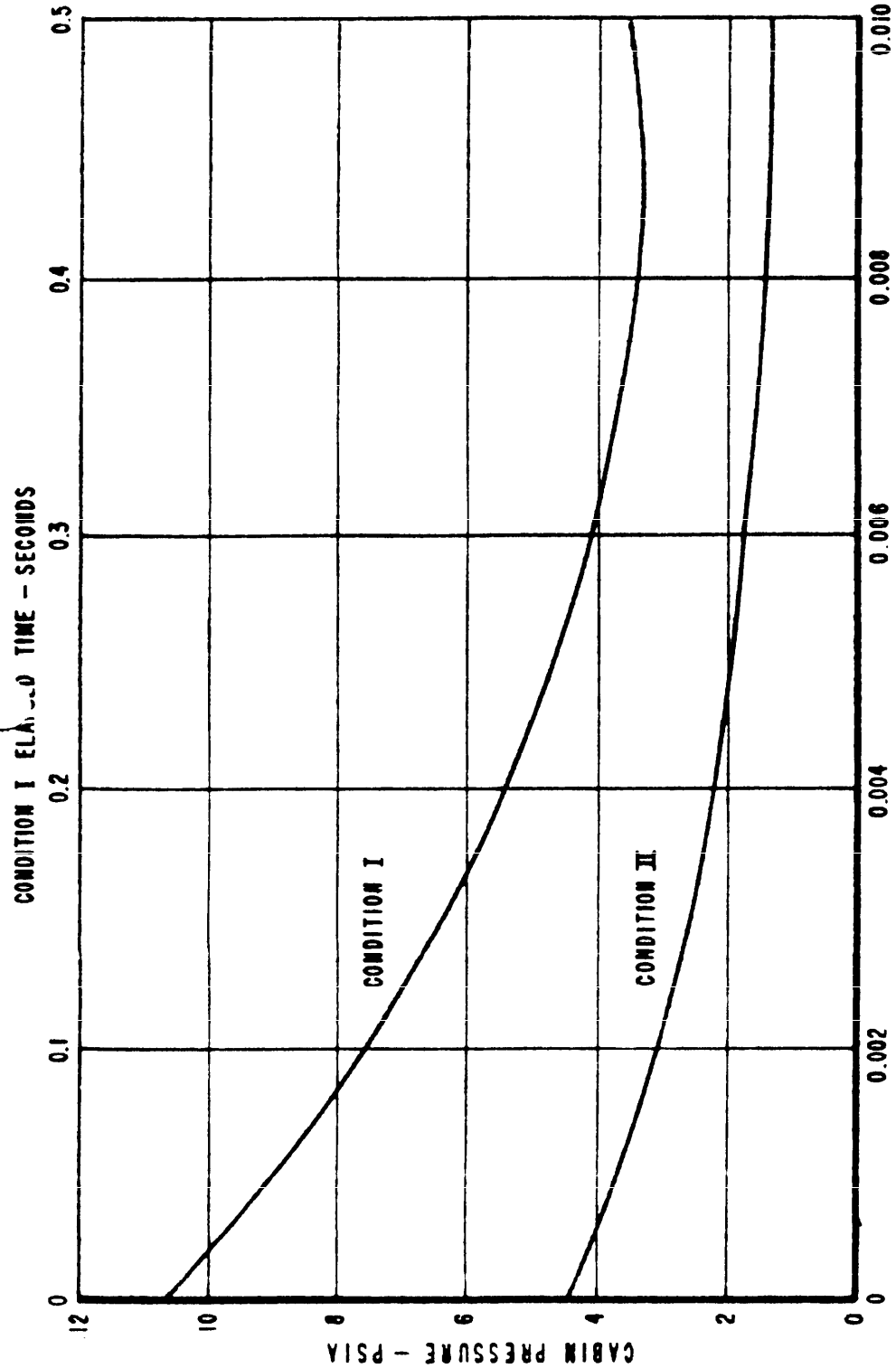


Figure 3. Cabin Pressure Decay During Explosive Decompression (Conditions I and II)

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1,000 cycles for each function shall be conducted over the entire range of the scale and shall cause the slip clutch to engage for at least two complete revolutions of the set knobs at both ends of the scale. Prior to the cycling test, the torque required to turn the set knobs shall be measured and recorded. Upon completion of the cycling test, a physical and functional operation inspection in accordance with 4.6.2.1, 4.6.2.2, and 4.6.5 (except that the abbreviated test schedule shall apply) shall be conducted to assure that binding, deformation, and malfunction are not evident. The torque required to turn the set knobs shall again be measured and shall not differ from the measurement recorded at the start of the cycling test by more than 2 inch-ounces.

4.6.25 Shock. At the completion of all other tests, one timer shall be subjected to a shock test in accordance with MIL-STD-810, method 516, procedure III, except that the severity of the shocks shall be 50g and the duration of each shock shall be 8.5 milliseconds. Upon completion of this test, the timer shall be subjected to the tests specified in 4.6.2.1, 4.6.2.2, and 4.6.5, except the abbreviated test schedule shall apply.

4.6.26 Reliability. The six timers specified in 4.4.1 shall be subjected to a reliability test to demonstrate a minimum acceptable MTBF of 1,000 hours to a 90 percent confidence level. The multiples of the MTBF shall be in accordance with table II. No preventive maintenance shall be performed. The total test time shall be computed in accordance with the following formula:

$$\text{Total test time per unit} = \frac{\text{Multiples of MTBF} \times \text{MTBF} \times \text{Discrimination Ratio}}{\text{No. of Test samples}}$$

or

$$\text{Time} = \frac{\text{MM} \times 1000 \times 1.5}{6}$$

TABLE II. Reliability Multiples of MTBF

No. of Failures	Multiples of MTBF
0	4.4
1	5.2
2	6.0
3	6.8
4	7.6
5	8.5
6	9.3

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4.6.26.1 Test equipment. Test equipment and fixtures shall include provisions for operation cycling, temperature cycling, and vibration.

4.6.26.2 Cycling. The timers shall be cycled at a rate of 2 to 30 cycles per hour throughout the reliability test. One cycle shall consist of timer reset, start, operation, and timeout. The timer setting shall be changed after each 100-hour performance check and shall vary to cover the timing range.

4.6.26.3 On-off cycling. During temperature cycling, the timer shall be turned off for the first hour of the room-temperature cycle.

4.6.26.4 Temperature cycling. The temperature of the test chamber shall be cycled as follows: Starting at room temperature, increase temperature to 71°C and stabilize for 2 hours; decrease temperature to -35°C and stabilize for 2 hours; increase temperature to room temperature and maintain for 2 hours; repeat.

4.6.26.5 Vibration cycling. Throughout this test, the timers shall be vibrated for a period of 10 minutes during each hour at any frequency between 20 and 60 Hertz at an applied acceleration of ±2g.

4.6.26.6 Performance check. At each 100-hour interval and at the end of reliability testing, the timers shall be checked for voltage drop and scale error (abbreviated test schedule shall apply). During each scale error check, if any test points exceed the allowable tolerances by 0.1 second, one malfunction shall be charged against this test.

4.6.26.7 Failure criteria. For an accept decision, each timer shall be operated to within 50 hours of its total time. If a timer fails, all testing shall stop and the failed unit examined to determine the cause of failure. The unit shall be repaired and testing of the unit continued. The procuring activity shall be immediately informed of the failure, what caused the failure, and the type of repair performed. A maximum of six failures will be permitted.

4.6.26.8 Reporting. The reliability test log shall be included in the test report specified in 4.4.2.

4.6.26.9 Longevity. Testing of two of the reliability test samples shall be continued for a total of 2,500 hours on each. A test log shall be submitted to the procuring activity.

4.7 Inspection of preparation for delivery. The inspection and testing of preparation for delivery shall be in accordance with section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A. Each timer shall be preserved in accordance with MIL-P-116, method IA-8 and packaged in accordance with MIL-STD-794.

5.1.2 Level C. Timers shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing, all levels. Timers preserved and packaged as specified in 5.1.1 and 5.1.2 shall be packed in accordance with MIL-STD-794.

5.3 Marking for shipment. Timers shall be marked for shipment in accordance with MIL-STD-129.

5.4 Reinspection date markings. Reinspection date markings shall be in accordance with the instructions of the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The precision timer covered by this specification is intended for use in bomber aircraft for timing bombing and navigation runs.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type of timer required (see 1.2)

c. Levels of packaging and packing required and reinspection date marking (see section 5).

6.2.1 Data. The contractor will submit the following to the preparing activity:

a. Three copies of the test report (see 4.4.2) covering compliance of the timer with all test data specified herein

b. Outline drawings of the timer covering dimensions, face plate layout, and other configuration details required by this specification

c. All test samples on which the qualification tests were performed

d. Test box or wiring harness for laboratory use.

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6.3 Definitions

6.3.1 Timing phase. The timing phase is the time from the start of the timer until time out (such as 25 minutes 00.0 second).

6.3.2 Face plate. The face plate is the integrally lighted face of the timer.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, Attn: ENFIP, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

6.5 Marginal indicia. Asterisks are not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.

Custodian:
Air Force - 11

Review activity:
Air Force - 82

Preparing Activity:
Air Force - 11

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